
Haoteng Zhao

Contact:

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Education

2018-2023: George Mason University, Fairfax VA, USA

Ph.D.: Department of Geography and Geoinformation Science, Earth System and Geoinformation Sciences

Dissertation: A data-driven intelligent decision-making model for irrigation scheduling

GAP:3.91/4.0

2015-2018: Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China

M.S.: Department of Data Management, major in Signal and Information Processing

Thesis: Research on quarry area monitoring and analyzing using Landsat-based long time series images

GPA: 3.39/4.0

2011-2015: Wuhan University, Wuhan, China.

B.S.: School of Remote Sensing and Information Engineering, major in Remote Sensing

Thesis: Using a dense matching method based on SGM to acquire point cloud

GPA: 3.44/4.0

Professional Experience

Science Systems and Applications, Inc.

Lanham, MD

Senior Research Scientist

Jul 2024 to present.

- Project: FFAR Metrics, Management, and Monitoring project
- Support USDA-ARS to develop remote sensing-based metrics of ecosystem health and response to climate and management.

USDA-ARS Hydrology and Remote Sensing Laboratory

Beltsville, MD

Postdoctoral Research Associate

Jul 2023 to present.

- Project: Data Fusion and Crop Phenology with High-Performance Computing (HPC)
- Mapped crop emergence through fusing multiple remote sensing data.
- Assessed Crop condition based on phenologically corrected vegetation index.
- Demonstrated phenologically corrected vegetation index performs better in yield prediction.
- Presented "Assessment of Crop Condition based on Phenologically Corrected Enhanced Vegetation Index Time Series from Harmonized Landsat Sentinel-2 (HLS) data" at American Geophysical Union 2023 in San Francisco, CA.
- Project: FFAR Metrics, Management, and Monitoring project
- Assessing various optical remote sensing data sources and parameters for tracking pasture changes due to weather and management.
- Evaluated remote sensing LAI products at Michigan and Oklahoma sites against ground observations.

Center for Spatial Information Science and Systems

Fairfax, VA

Graduate Research Assistant

Sep 2018 to May 2023.

- Project: Crop-CASMA: A Web-based Geospatial Application for Crop Condition and Soil Moisture Analytics
- Designed & developed the [Crop-CASMA portal](#) based on React framework.
- Developed the web analysis tools for online geospatial data processing.

- Demonstrated the impact of soil moisture on crop health based on time series analysis using Google Earth Engine
- Project: “WaterSmart”: A Cyberinfrastructure-Based Integrated Agro-Geoinformatic Decision-Support Web Service System to Facilitate Informed Irrigation Decision-Making
- Created soil moisture, ET, soil property data products for the state Nebraska using HRLDAS and USDA-NCSS soil survey data (SSURGO)
- Automated the harvest, generating, process, and publish of hourly/daily data products.
- Deployed Web Map Service and Web Processing Service for data publishing and analyzing.
- Designed, developed, maintained the <https://geobrain.csiss.gmu.edu/watersmartport/web/> based on React framework.
- Maintained and enhanced the <https://geobrain.csiss.gmu.edu/ids/>
- Presented “Validation and Calibration of HRLDAS Soil Moisture Products in Nebraska” at Agro-Geoinformatics 2022 in Quebec City, Canada
- Proposed a deep reinforcement learning approach for optimizing field-level irrigation decision making.

Institute of Remote Sensing and Digital Earth

Beijing, China

Graduate Research Assistant

Sep 2016 to Jun2018

- Project: Quarry Change Survey of Northern China in the Past Three Decades
- Developed a machine learning method for the classification of Landsat long time series data.
- Project: Using Active and Passive Collaborative Monitoring Method for Land Cover Change Detection
- Accomplished the method of dynamic change detection in Hulun Lake Nature Reserve
- Generated a quick result of post-disaster assessment after Jiuzhaigou earthquake.
- Attend China’s National Science Week exhibition as an exhibitor in May 2017
- Narrated the demo system about satellite data receiving and displaying for visitors.
- Project: Emergency Post-Disaster Change Detection Using Multi-Sourced Remote Sensing Data
- Implemented a SGM based algorithm which can produce digital elevation models from push-broom images.
- Conducted a DEM differencing method for monitoring topographic change.

Honors and Awards

- Outstanding PhD Student at Department of Geography and Geoinformation Science, George Mason University, 2023
- The First-Prize Award of Agro-Geoinformatics 2022 Best Student Paper Competition, 2022
- Outstanding Graduate Student Award, Institute of Remote Sensing and Digital Earth, 2018
- Pacemaker to Merit Student, Institute of Remote Sensing and Digital Earth, 2018
- Excellent Students Awards of the first graduate year, University of Chinese Academy of Sciences, 2016

Selected Publication

- **Zhao, H.**, Gao, F., Anderson, M., Cirone, R. and Chang, G.J., 2025. Improving crop condition monitoring using phenologically aligned vegetation index anomalies—A case study in central Iowa. *International Journal of Applied Earth Observation and Geoinformation*, 139, p.104526.
- **Zhao, H.**, Di, L., Guo, L., Zhang, C. and Lin, L., 2023. An Automated Data-Driven Irrigation Scheduling Approach Using Model Simulated Soil Moisture and Evapotranspiration. *Sustainability*, 15(17), p.12908.
- **Zhao, H.**, Di, L., Guo, L., Li, L., Zhang, C., Yu, E. and Li, H., 2023, July. Optimizing Irrigation Scheduling Using Deep Reinforcement Learning. In *2023 11th International Conference on Agro-Geoinformatics (Agro-Geoinformatics)* (pp.

1-4). IEEE.

- **Zhao, H.**, Di, L., Yu, E., Guo, L., Li, L., Zhang, C. and Li, H., 2023, July. A Review of Scientific Irrigation Scheduling Methods. In *2023 11th International Conference on Agro-Geoinformatics (Agro-Geoinformatics)* (pp. 1-4). IEEE.
- **Zhao, H.**, Di, L., Sun, Z., Yu, E., Zhang, C. and Lin, L., 2022, July. Validation and Calibration of HRLDAS Soil Moisture Products in Nebraska. In *2022 10th International Conference on Agro-geoinformatics (Agro-Geoinformatics)* (pp. 1-4). IEEE.
- **Zhao, H.**, Di, L. and Sun, Z., 2022. WaterSmart-GIS: A Web Application of a Data Assimilation Model to Support Irrigation Research and Decision Making. *ISPRS International Journal of Geo-Information*, 11(5), p.271.
- Zhang, C., Yang, Z., **Zhao, H.**, Sun, Z., Di, L., Bindlish, R., Liu, P.W., Colliander, A., Mueller, R., Crow, W. and Reichle, R.H., 2022. Crop-CASMA: A web geoprocessing and map service based architecture and implementation for serving soil moisture and crop vegetation condition data over US Cropland. *International Journal of Applied Earth Observation and Geoinformation*, 112, p.102902.
- **Zhao, H.**, Di, L., Sun, Z., Hao, P., Yu, E., Zhang, C. and Lin, L., 2021, July. Impacts of Soil Moisture on Crop Health: A Remote Sensing Perspective. In *2021 9th International Conference on Agro-Geoinformatics (Agro-Geoinformatics)* (pp. 1-4). IEEE.
- Zhang, C., Di, L., Hao, P., Yang, Z., Lin, L., **Zhao, H.** and Guo, L., 2021. Rapid in-season mapping of corn and soybeans using machine-learned trusted pixels from Cropland Data Layer. *International Journal of Applied Earth Observation and Geoinformation*, 102, p.102374.
- Yang, Z., Zhang, C., **Zhao, H.**, Sun, Z., Bindlish, R., Liu, P.W., Colliander, A., Mueller, R., Di, L., Crow, W. and Reichle, R.H., 2021, July. Crop-CASMA-a web GIS tool for cropland soil moisture monitoring and assessment based on SMAP data. In *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS* (pp. 6315-6318). IEEE.
- **Zhao, H.**, Ma, Y., Chen, F., Liu, J., Jiang, L., Yao, W. and Yang, J., 2018. Monitoring quarry area with landsat long time-series for socioeconomic study. *Remote Sensing*, 10(4), p.517.

Skills and Expertise

Programming languages: Python, Java, JavaScript, C++/C, Shell, R

Machine Learning: Keras, Tensorflow, PyTorch, Scikit-learn

Geospatial: Quantum GIS, ArcGIS, GeoServer, OpenLayers, ENVI, Erdas, eCognition, PostGIS

General: Node.js, React, Flask, Bootstrap, MySQL, Spring Boot, PostgreSQL, Eclipse, GitHub, D3.js, MongoDB, Neo4j

Teaching

GGIS 773: Interoperability of Geographic Information Systems

Department of Geography and Geoinformation Science, George Mason University

Guest Lecturer

Spring 2025

IEEE GRSS Washington, DC & Northern VA Chapter Virtual Seminar

(Lecture topic: A Data-Driven Intelligent Decision-Making Model for Irrigation Scheduling)

IEEE Geoscience and Remote Sensing Society

Guest Lecturer

2023

Related Experience

Professional Societies and Organizations

Member, AGU

2023-present

Student Member, AGU

2019-2023

Student Member, AAG

2018-2019

Journal/Conference Reviewer

Agronomy, MDPI	2022-2024
Agriculture, MDPI	2022-2024
Crops, MDPI	2022
Plants, MDPI	2022-2023
Remote Sensing, MDPI	2022-2024
Sustainability, MDPI	2023
Electronics, MDPI	2023
Applied Sciences, MDPI	2023
Atmosphere, MDPI	2023
Sensors, MDPI	2023-2024
Environmental Science and Pollution Research, Springer	2023
Urban Water Journal, Taylor & Francis	2023
International Journal of Digital Earth, Taylor & Francis	2024-2025
Computers and Electronics in Agriculture, Elsevier	2024
International Conference on Agro-Geoinformatics	2019-2023
Scientific Reports, Nature	2024
International Journal of Applied Earth Observation and Geoinformation	2025